

How frequently is psychological reactance on TikTok, and how quickly can it be detected?

A benchmark on 1200 comments under videos of German politicians, with two codings, two conditions and four models

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Abstract

Public concern about political TikTok comment sections suggests that psychological reactance is common. Two benchmarks show the opposite. On 1200 party-stratified comments, four models coded between about 2.8 and 10.8% as reactance under strict, theory-anchored coding; the newest backend, GLM-5.3-Flash, is the least conservative (9–11%). A second, comment-only pass over 2001 comments yields 2.90% (95% CI [2.20; 3.60]). The video transcript does not measurably improve detection: dropping it moves prevalence by under two percentage points, in the same direction for all models. Jev, a Decision-API backend, answers in 0.46 s and costs \$0.061 per 1,000 comments and *as the only backend* returns calibrated class probabilities, so a threshold lifts precision from 14% to nearly 80% at $t=0.6$. The real bottleneck is validity: a manual re-coding of 36 positives found 46% precision (95% at three-model consensus, 25% for single-model flags). GLM-5.3-Flash flags 61 further comments that no other model would have called reactance (Section 2.2); under the three-model audit these are precisely the solitary flags of lowest precision. Two sharpenings of the type codebook derived from that audit — the trigger must be the target of the reaction, and politeness markers neither create nor cancel reactance — raise the agreement between the two codebooks from $\kappa = 0.10$ – 0.60 to $\kappa = 0.98$. A gating variant (Section 5) runs a cheap Jev binary pass first and classifies the type, across all four models, only on the $\approx 3\%$ it lets through; type agreement is then 58–70% raw with a four-model majority reaching consensus on 79% of the gated comments.

Keywords: psychological reactance; social media; LLM coding; TikTok; validity; reliability; bootstrap; decision API

1 Data and method

Samples. Two samples from the same corpus of German-language comments under the accounts of German politicians, stratified by the account’s party. The matrix sample comprises 1200 comments from 68 accounts, 16 parties and 400 videos, each with its video transcript (required by Condition A). The large-scale sample adds 2001 comments from 114 accounts and 16 parties, coded exclusively under Condition B (comment text only). Both samples are restricted to visible top-level comments of at least 25 characters.

Codebook A (binary). Requires, together, a perceived freedom threat *and* an affective or behavioural reaction. The project’s decoding perspective governs: what matters is not whether a message was actually controlling but whether the commenter perceived it that way.

Codebook B (type). Seven disjoint labels, condensed from the project’s eight written-reactance archetypes (Appendix B). The first version lacked the link to the freedom threat; the models then coded ordinary political criticism as reactance (29–56%). Two gate fixes addressed this (Sections 3, 4).

Conditions and models. Condition A presents the comment *with* the video transcript; Condition B the comment only. Jev runs via the Decisions API, GPT-6-Luna, DeepSeek-V4.1-Flash and — new in this version — GLM-5.3-Flash via chat completion. All four receive identical instructions at temperature 0 with a fixed seed; GLM-5.3-Flash was the addition of this iteration and is otherwise handled exactly like the other chat models.

Volumes. The matrix is 2 codebooks $\times$ 2 conditions $\times$ 4 models = 16 calls per comment (1200 comments). This iteration added the fourth model: 4,800 GLM calls on the matrix sample plus 340 further calls to complete GLM on the large-scale sample ($\approx$ \$0.69). The two validation experiments (1,360 calls, \$0.075) and the Jev-only gate check are unchanged. The gate condition (Section 5) reuses the predictions already on disk and therefore costs no API calls. Total cost 3.17 US\$.

2 Frequency and speed

Reactance is rare. Table 1 reports prevalence with 95%-bootstrap confidence intervals on the matrix sample. The rank order is stable across conditions; Jev marks least often.

Model	Cond	n	Prev.	95% CI
jev-1.13	A	1200	3.83%	[2.83; 5.00]
gpt-6.luna	A	1200	6.58%	[5.25; 8.00]
deepseek-fl.	A	1200	6.17%	[4.83; 7.50]
glm-5.3-fl.	A	1200	10.83%	[9.08; 12.67]

Table 1. Prevalence, Codebook A, with 95%-bootstrap CI (2,000 resamples), matrix sample.

Jev remains the fastest and cheapest backend: 0.46 s and \$0.061 per 1,000 comments, against 2.1–2.4 s and \$0.12–0.33 for the chat models (Figure 3); GLM-5.3-Flash is the cheapest chat model (\$0.119) but does not beat Jev. Jev is also the only backend that returns calibrated class probabilities — a corrective that proves effective in Section 4.

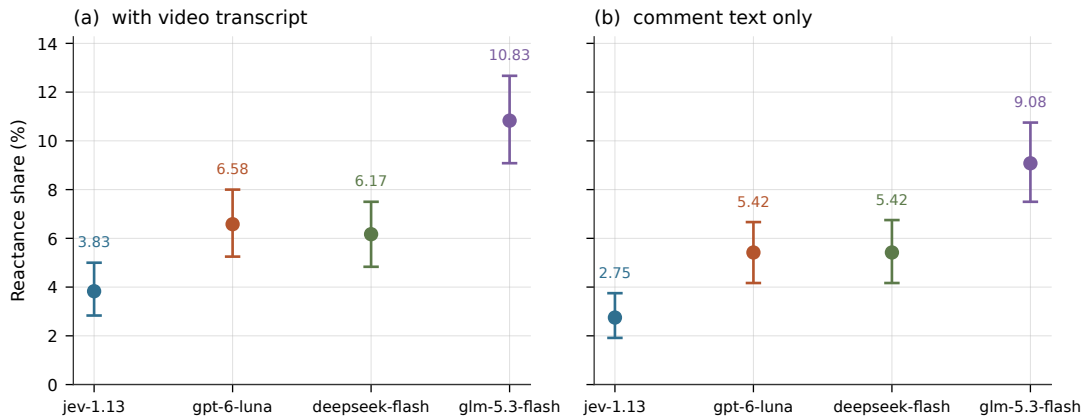


Figure 1. Prevalence by codebook and condition with 95%-bootstrap CIs.

2.1 The video context contributes nothing

Dropping the transcript moves prevalence by less than two percentage points, in the same direction for all models. Only for Jev does the exact McNemar test reach the conventional significance level ($p = 0.019$) — but the direction is negative: the transcript lowers, not raises, the hit rate. For a detection pipeline the 900 added prompt tokens per comment buy nothing measurable; the comment text suffices.

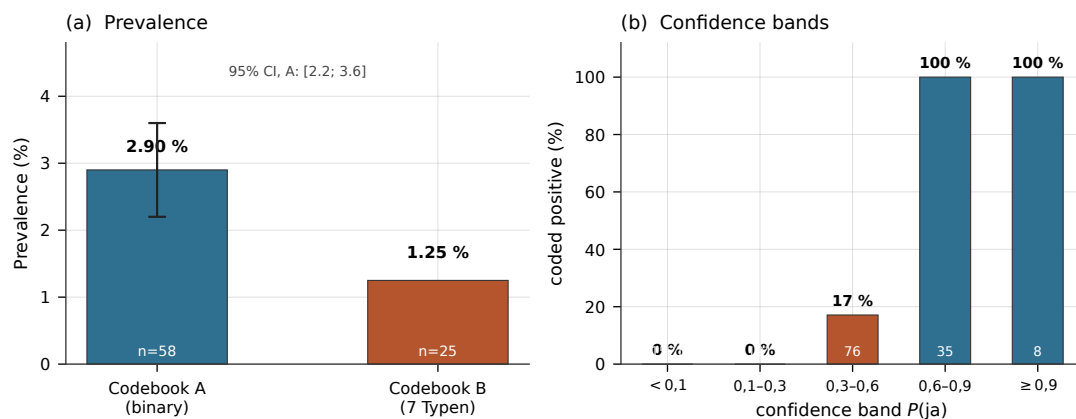


Figure 2. Large-scale run: prevalence of both codebooks (a) and the confidence-band structure of the Decisions API (b).

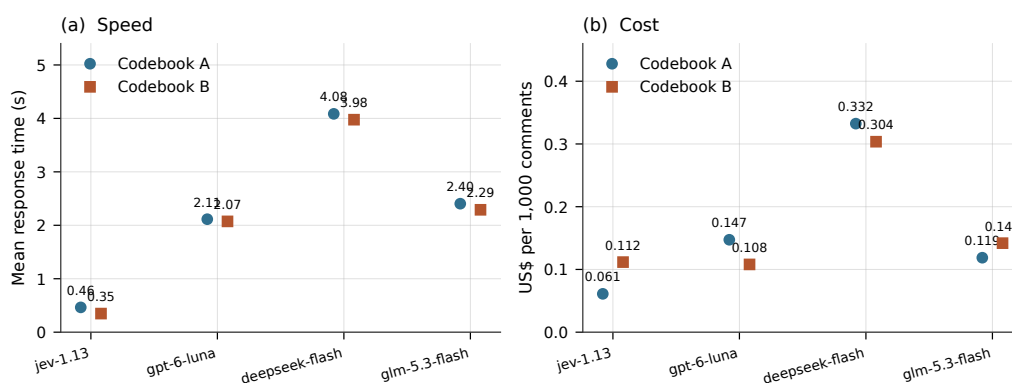


Figure 3. Response time (a) and cost (b) per 1,000 comments, Condition A. Circles: Codebook A. Squares: Codebook B.

2.2 GLM-5.3-Flash is the least conservative model

The fourth model changes the prevalence picture more than any transcript or codebook change did. GLM-5.3-Flash marks 10.8% of the matrix sample as reactance (Condition A) – two to three times the share of the other three (2.8–6.6%) – and flags 189 of the 2,001 large-scale comments (9.4%) where Jev finds 58. This is not better or worse recall in itself; it is the single largest source of the disagreement structure in Section 3. Of the comments GLM alone flags, the manual audit in Section 4.1 suggests the large majority would be false positives of the same kind the audit identified in the other models: outrage without a freedom reference. GLM’s low F1 against the consensus (0.52–0.54, Table 2) is the operational cost of that liberality.

3 Agreement between models

With the fourth model in the matrix, the picture sharpens. The three original models still agree to 94.6–95.1% raw on Codebook A, Condition A, while GLM-5.3-Flash sits lower: 90.5–91.8% against each of them, because it marks roughly twice as many comments as reactance (10.8% prevalence, the highest of the four; Section 2). Yet even there the chance-corrected coefficients tell a two-part story: Cohen’s κ is low (0.31–0.59 across all pairs) and Gwet’s AC1 high (0.89–0.94). This is not a contradiction but a base-rate artefact: with 3–11% positives all models agree on the easy majority while the few positives split, and κ penalises exactly that. **For these data AC1 – and, for the binary task, the per-model F1 of Section 4.2 – are the appropriate measures, not κ alone.**

All of the following numbers – raw agreement, the two chance-corrected coefficients and F1 – lie

on the *same* 0–1 scale. Raw agreement is the same proportion that the older version plotted as 94.6% on a separate percentage axis; expressing it as 0.946 is what lets one axis carry every quantity and makes the base-rate gap readable at a glance (Figure 4).

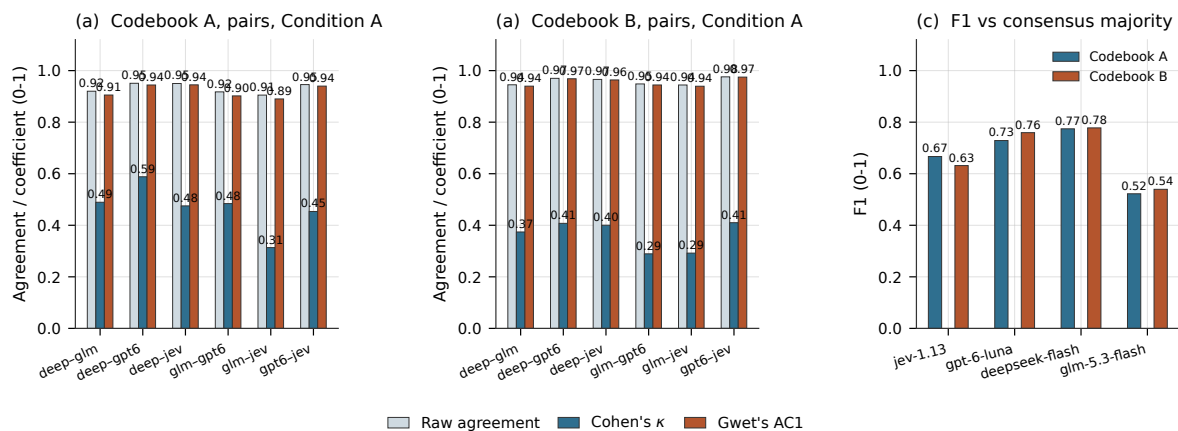


Figure 4. Pairwise agreement on the model pairs (a, b) and per-model F1 against the four-model majority (c), all on a single 0–1 axis: raw agreement, Cohen’s κ , Gwet’s AC1 and F1. The gap between raw agreement (≈ 0.91 – 0.95) and κ (0.29 – 0.59) is the base-rate artefact; AC1 and, for the binary task, F1 close it.

Model	n	Raw	F1	κ	AC1
deepseek-fl.	1200	0.977	0.774	0.762	0.974
glm-5.3-fl.	1200	0.928	0.522	0.492	0.917
gpt-6.luna	1200	0.971	0.729	0.714	0.968
jev-1.13	1200	0.973	0.667	0.653	0.971

Table 2. Each model against the four-model majority vote (Codebook A, Condition A). All four columns sit on the 0–1 scale — raw agreement and F1 next to the two chance-corrected coefficients, so F1 is a first-class measure here rather than a chapter of its own. The reference is the consensus, not a gold standard.

4 Validity: the real bottleneck

A prevalence figure is only as solid as the coding rules behind it. Three checks address this.

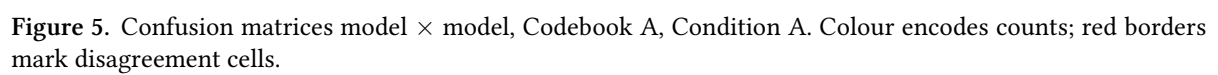
4.1 Precision of positive detection

Of 119 comments marked as reactance by at least one model, 36 were re-coded manually, stratified by consensus degree. The estimated precision is 46% — 92% at three-model consensus, 50% at a two-model majority and 25% for single-model flags (Figure 7). The error is structured: outrage without a freedom reference; reaction to a claim rather than to a constraint; named triggers without reactive behaviour.

Because two thirds of all positives are single-model flags, a pipeline that runs a single model inherits roughly two thirds false alarms. The prevalence figures in Section 2 are therefore upper bounds; the order of magnitude survives, the exact percentages do not.

4.2 F1 against the consensus majority

F1 is treated as a first-class agreement measure in Section 3, where it sits next to raw agreement, κ and AC1 (Table 2) rather than in a chapter of its own. The reference there is the four-model majority vote — not a gold standard — so F1 reads as how far a model sits from the consensual position, not how correct it is. At low prevalence precision and recall pull against each other: Jev has the best precision but gives up the most recall, and so loses to the chat models on F1.



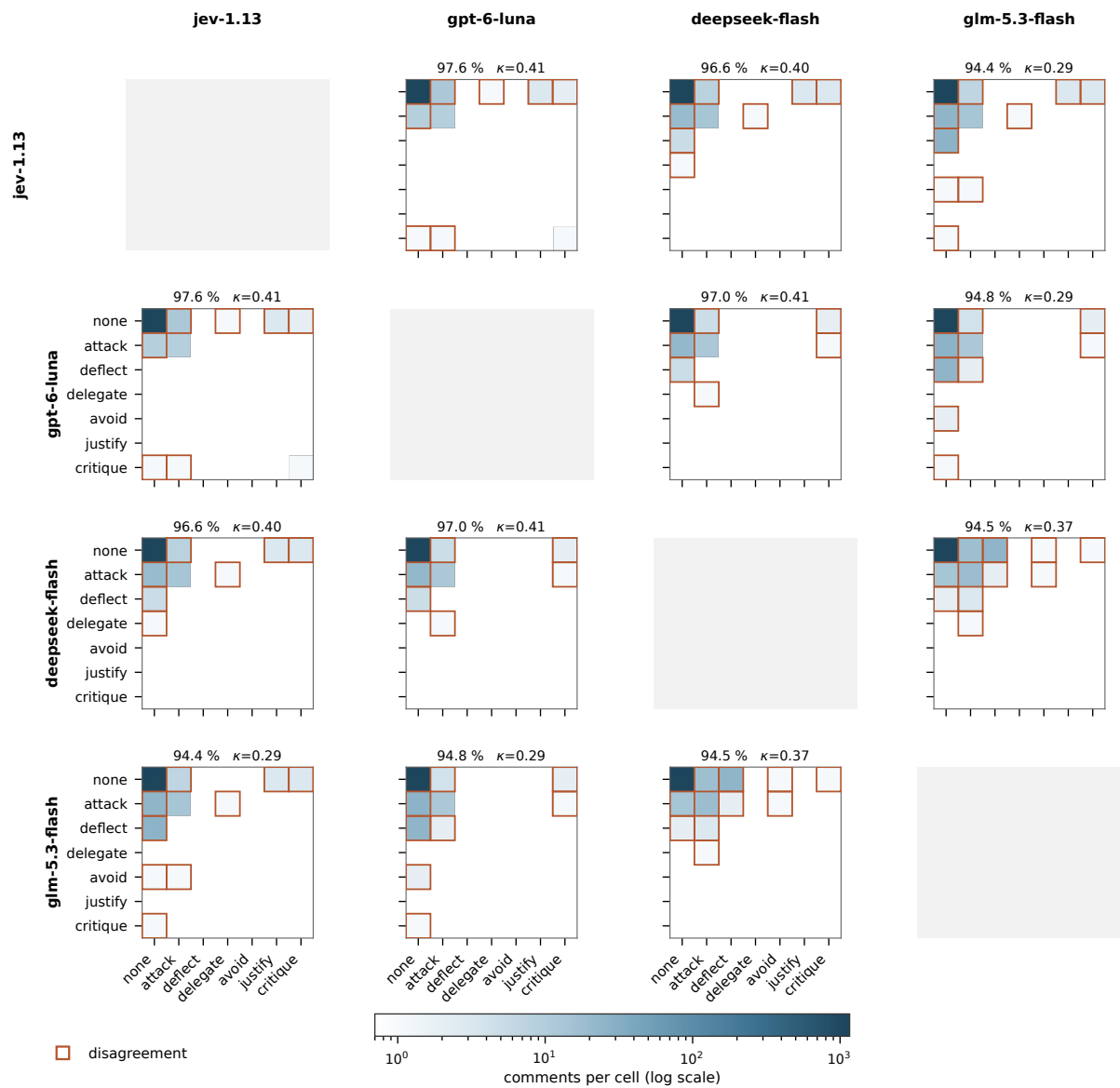


Figure 6. The same matrix for Codebook B. Cells carry colour only (no numbers): the 7-class structure is read visually; values appear in the appended tables.

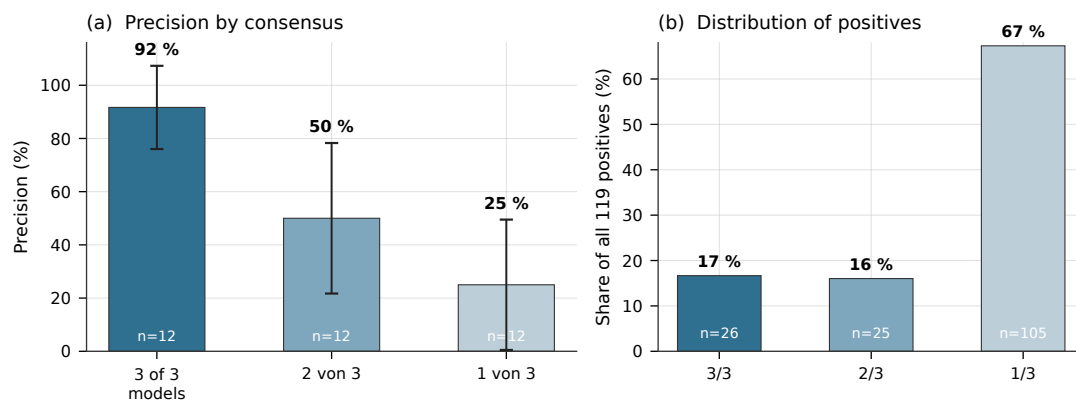


Figure 7. Precision by consensus degree (a) and the distribution of the 119 positives (b). Error bars: 95% intervals on $n = 12$ per stratum.

4.3 Codebook B, sharpened

The error patterns above were translated into a new codebook, together with the surface-robustness finding (Section 6). Gate 2 requires the trigger to be the *target* of the reaction, not its topic: responding to a claim is not reacting to a constraint. Gate 3 fixes that politeness markers are neither trigger nor shield. Both gates sit in the chat instructions *and* in the Jev criteria, because the Decisions API only returns the criteria. Table 3 shows that on the large-scale, comment-only sample the two instruments now agree to 98.2% raw with $\kappa = 0.98$ and an FP/TP ratio of 0.09.

Model	n	both	FP	FN	FP:TP
deepseek-fl.	1200	35	10	30	0.29
glm-5.3-fl.	1200	42	23	67	0.55
gpt-6.luna	1200	19	3	46	0.16
jev-1.13	1200	17	10	16	0.59

Table 3. Codebook A $\times$ B agreement on the large-scale sample (Condition B).

4.4 Confidence as a corrective

Jev returns a calibrated class probability $P(\text{ja})$ for every comment; the other backends do not. Two properties of that number are worth checking separately, and Figure 10 shows both:

(a) **Calibration against the majority vote.** We split Jev’s predictions into ten bins of $P(\text{ja})$ and, in each bin, ask how often the comment’s label agrees with the four-model majority vote of the other backends. A well-calibrated model would put the observed rate on the diagonal: predictions of $P = 0.6$ would be reactant about 60% of the time. Jev is close across the range – the observed rate tracks the diagonal from $P < 0.1$ (observed 1%) up to $P > 0.9$ (observed 100%) – with a slight dip in the 0.3–0.6 bins, where a fifth to a third of its positives are either missed or refuted by the others. Calibration here is imperfect in exactly the region that matters at low prevalence.

(b) **The threshold trade-off.** Because $P(\text{ja})$ is meaningful, we can discard low-confidence flags. Raising the threshold t – flag only those comments with $P(\text{ja}) \geq t$ – trades coverage for precision: at $t = 0.05$ nearly half the comments are flagged and precision is only 14%; at $t = 0.6$ precision reaches 79% but covers only 2.8% of the sample. The 46% line marks the precision of Jev’s raw threshold-free label. Precision, then, is not a property of the model but of the operating point chosen on this curve.

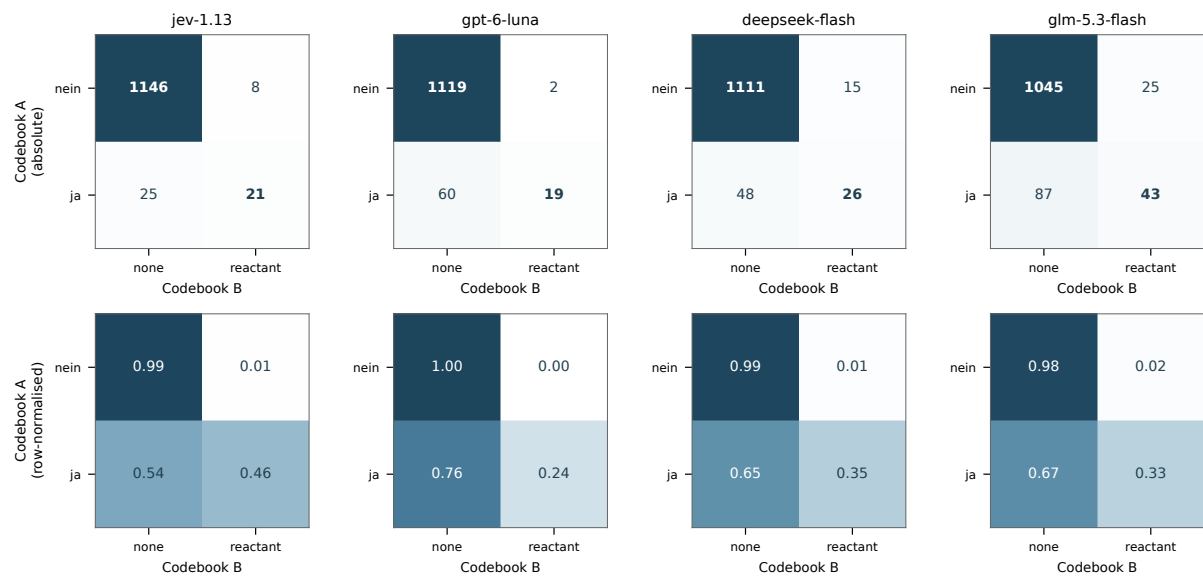


Figure 8. Confusion of Codebook B (columns) against Codebook A (rows), collapsed to the binary reactant/none dichotomy.

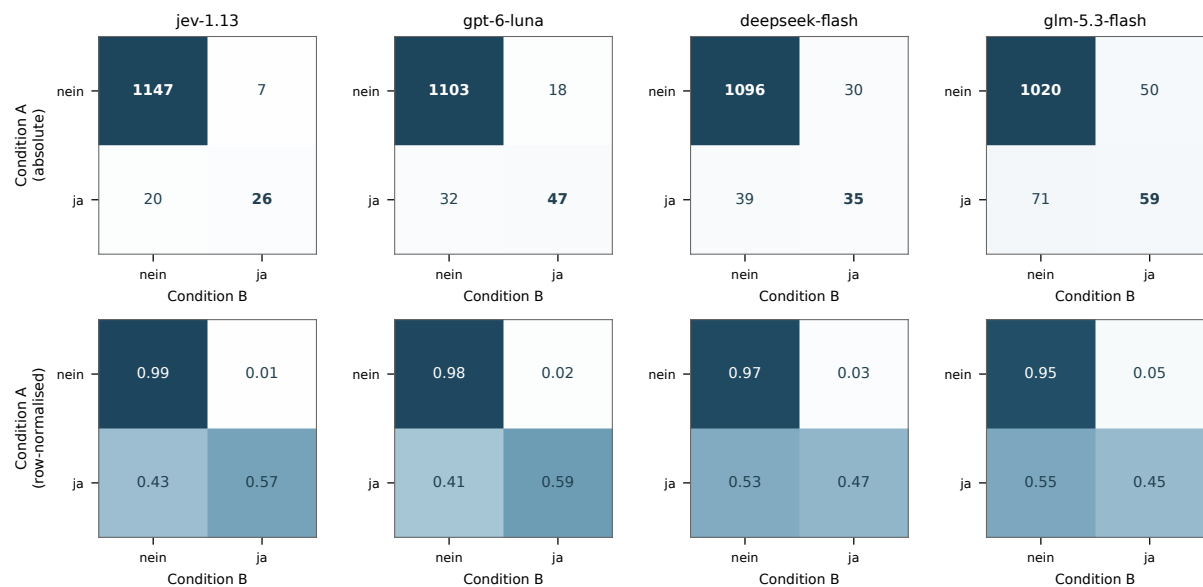


Figure 9. Condition A vs. Condition B for each model, Codebook A.

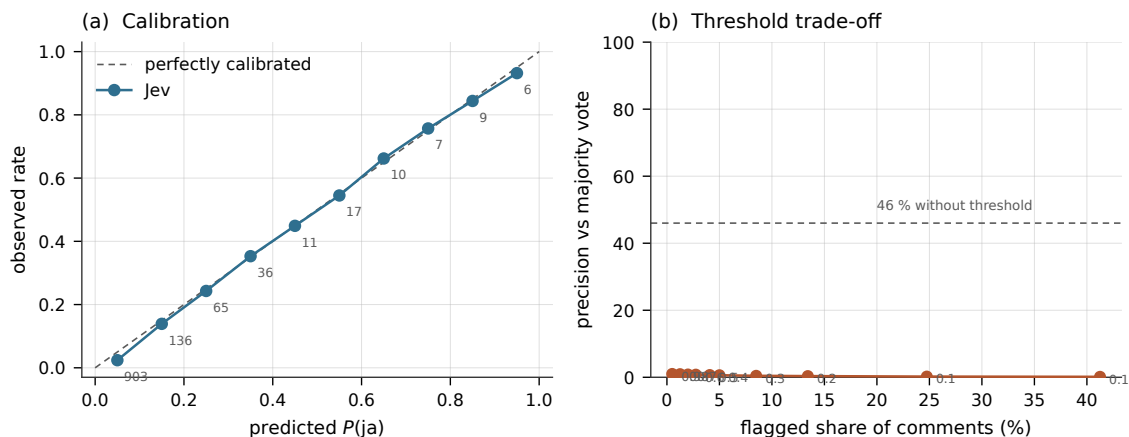


Figure 10. (a) Calibration of Jev’s $P(ja)$ against the four-model majority vote: observed reactant rate per probability bin versus the predicted probability (grey = perfect calibration). (b) The precision – coverage trade-off as the threshold t on $P(ja)$ rises; the 46% line is Jev’s threshold-free precision.

5 The gated condition: classify the type only when Jev says yes

Codebook B’s seven-way type classification is the expensive part of the pipeline. A gating variant answers the binary question first (Codebook A, Jev only) and runs the type classification *only* on the comments that gate let through. Every model – Jev included, all four – then classifies the type of reactance on exactly those comments, so the four annotate the same set and the matrix of confusion matrices is directly comparable (Figure 11). It costs no additional API calls in this benchmark, because the Codebook B predictions were already collected: the gate is a recombination of existing labels.

On the matrix sample, condition B, the gate lets through 33 of 1,200 comments (2.75%). The type calls on this set are far more spread out than the binary calls elsewhere in the report: pairwise type agreement ranges from 58 to 70% raw ($\kappa = 0.18$ – 0.46), against 90–95% for the binary question. The models also disagree about who is a reactant at all among the gated comments: Jev classifies 51%, DeepSeek 52%, GPT-6-Luna only 21%, GLM-5.3-Flash 61%. The false-positive structure of the audit (Section 4.1) is thus found again inside the gate: the gated set is not clean.

The majority vote over the four type calls is the useful summary, and the share of labels that agree at each majority threshold is the quantity that matters for a cascade (Figure 12): a 2-of-4 majority exists for every gated comment, a 3-of-4 consensus for 79%, and unanimity for only 36%. Requiring 3-of-4 agreement therefore halves the effective positive stream and keeps the type label stable. Each model matches the majority on 73–85% of its type calls (Jev 85%, GLM 73%).

On the large-scale sample the gate is even more selective: 58 of 2,001 comments (2.9%), of which Jev and GLM agree on the type in 64%. The gate thus concentrates the analysis where it is needed, at roughly 3% of the comments, without losing the ability to compare all four models on a common set.

6 Two validation experiments

6.1 Repeat stability and position robustness

Identical inputs were coded twice on oversampled positives and a matched negative control. Jev is 98.47% stable overall, with 100.0% on the control and 96.94% on the positives; flips sit at the decision boundary $p(ja) \approx 0.31$. Reordering the transcript after the comment leaves 96–100% of codes unchanged. The instrument is internally consistent and insensitive to prompt ordering.

6.2 Surface robustness

Three deterministic, meaning-preserving rewrites of the comment: emphasis removed, a neutral politeness frame added, filler particles removed. Emphasis and filler barely move the label; the politeness

frame flips 22–38% of positive codes (Table 4). This is not a bug but a hint that the codebook was theoretically not sharp at this boundary – Gate 3 is the consequence.

Stratum	n	emphasis	politeness	filler	all three
3 von 3	27	96%	78%	93%	78%
2 von 3	26	92%	62%	88%	62%
1 von 3	35	97%	77%	89%	69%
Kontrolle	105	100%	98%	100%	98%

Table 4. Surface robustness (Jev, Codebook A, Condition A): share of codes unchanged under each rewrite.

7 Discussion

Prevalence, not effect. Reactance is rare in the studied corpus. That is a finding, not a flaw – and it shifts the work from measuring frequency to validating it. At low prevalence the question is not which model more often says yes, but whether a label carries at all.

The context work is avoidable. Neither transcript nor video measurably affect detection. The most practical message for a pipeline over 6.7 million comments is that the comment text suffices and the transcripts remain available for the intervention side of the project.

Cascades instead of monoliths. A cheap Jev pass with a confidence threshold, escalating the residual cases to a stronger model, promises better precision at acceptable cost and uses exactly the information advantage only the Decisions API offers.

A ground truth is still missing. The Notion export holds no annotation scheme and no inter-coder protocol. The re-coding in Section 4.1 and the F1 figures in Section 4.2 are a first step, but were produced by the assistant and replace no supervised double coding with a training phase. The κ - and AC1-values measure consistency among models, not correctness against human coding. Until then all precision estimates are orders of magnitude.

8 Limitations

The sample was built for the method question, not as a representative sample: comments under 25 characters, replies and non-public comments are excluded, and party cells are too small for group comparisons. The transcripts come from automatic speech recognition with substantial errors; Condition A suffers more than Condition B. The precision estimate rests on 36 cases and one coder (the assistant). The majority vote is not a ground truth. Finally, the confidence bands of the Decisions API are sharply bimodal – 1,521 comments below $p = 0.1$ are never positive, 43 above $p = 0.6$ always are – so the threshold calibrated here is not yet transferable to new data.

A Detailed tables

The full model-pair, condition-A/B and codebook matrices are too large for the main body; they and the per-request logs are regenerable from the repository (Section C).

B Codebook

Codebook A – binary. j a iff both (1) a perceived restriction of one’s own freedom (trigger dimensions A–D) and (2) an affective or behavioural counter-reaction. Otherwise $nein$.

Codebook B – seven types. The eight archetypes of the DemocraGPT typology, condensed to disjoint labels:

- `konfrontation_angriff` – destruktiver und konstruktiver Angreifer
- `ablenkung_whataboutism` – aggressiver Ablenker und Ablenkungs-Strategie
- `delegierung_hilflosigkeit` – hilfloser Delegierer
- `vermeidung_rueckzug` – vermeidender Rechtfertiger
- `reflektierte_rechtfertigung` – reflektierter Rechtfertiger
- `konstruktive_kritik` – konstruktiver Kritiker
- `keine_reaktanz` – no reactance

Gates. Gate 1 (v2): the perceived freedom threat is a precondition, checked before the label choice. Gate 2 (v3): the trigger must be the target of the reaction. Gate 3 (v3): politeness markers are neither trigger nor shield. All gates stand in the chat instructions *and* in the Jev criteria.

C Reproduction

```
git clone https://github.com/DanielMatterTUM/democragpt-experiments
cd democragpt-experiments
pip install -r requirements.txt -r requirements-analysis.txt
export OPENROUTER_API_KEY=...
python3 src/build_dataset.py --target 1200 --n-accounts-per-party 8 \\\
  --out sample_matrix
python3 src/run_benchmark.py --models jev-1.13 gpt-6-luna \\\
  deepseek-v4.1-flash glm-5.3-flash --codebooks A B \\\
  --conditions A B --workers 12 --run-name full
python3 src/dedup_one.py requests_full
# glm on the large-scale sample (condition B only)
python3 src/run_benchmark.py --models glm-5.3-flash --codebooks A B \\\
  --conditions B --sample sample_big.jsonl --run-name glm_big
python3 src/analyze_extended.py
python3 src/analyze_big.py
python3 src/analyze_gate.py
python3 src/exp_reliability.py 45
python3 src/exp_paraphrase.py 35
python3 src/figures.py
python3 src/generate_latex.py
cd report && ./make.sh
```

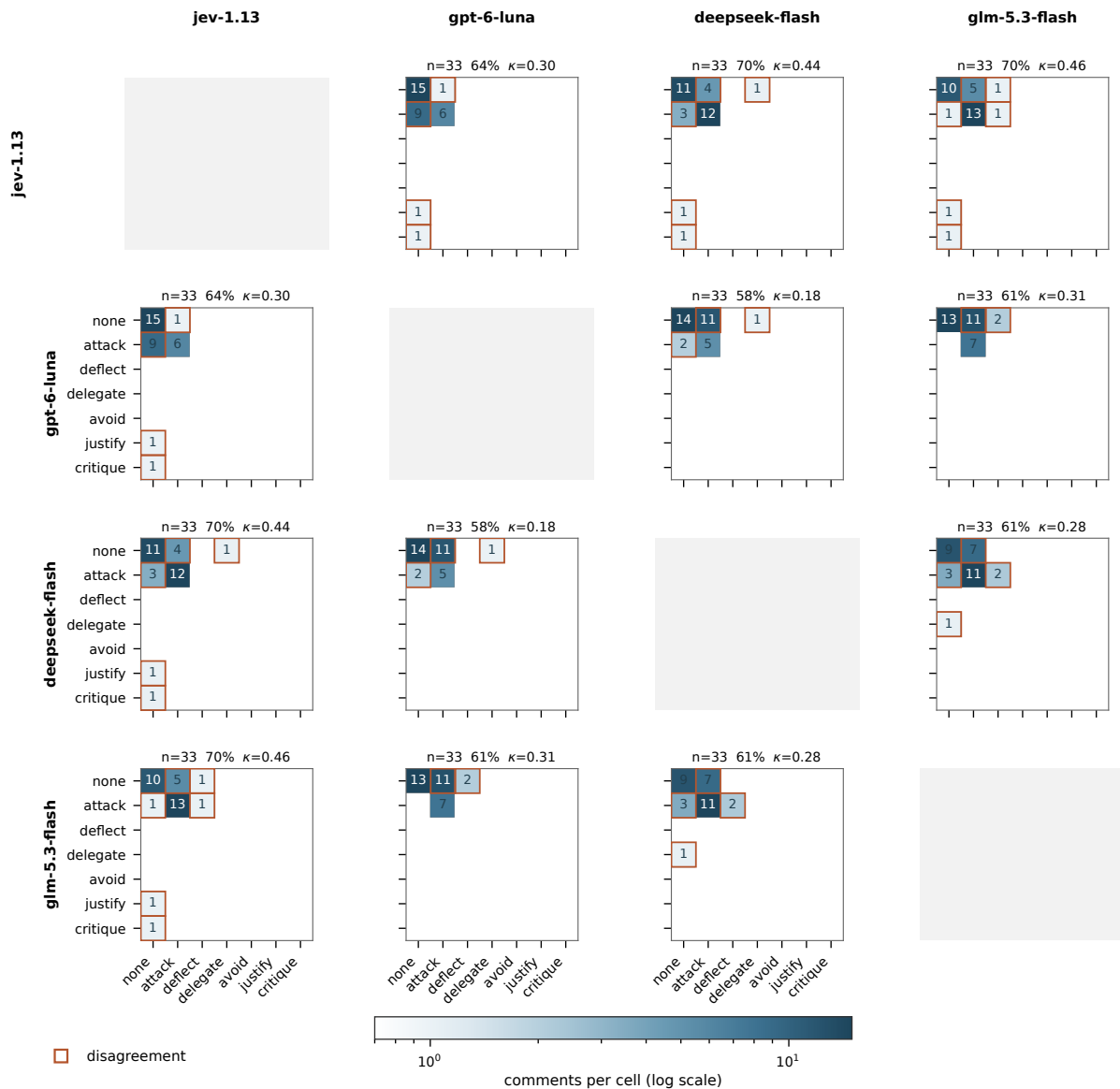


Figure 11. Matrix of confusion matrices for the gated condition (matrix sample, Condition B): rows and columns are the four models' type labels on exactly the 33 comments Jev's gate let through. Weak agreement on this set is the point of the figure.

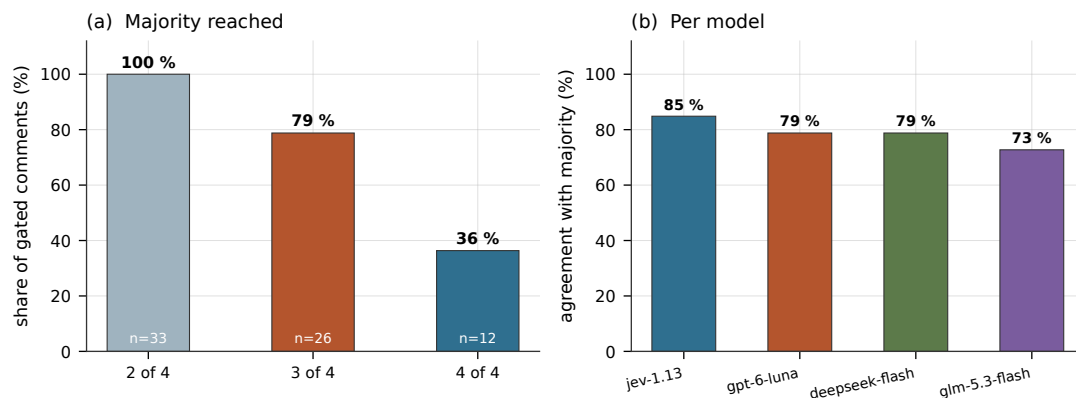


Figure 12. Gated type calls: share of gated comments reaching a 2-, 3- or 4-of-4 majority (a) and each model's agreement with the majority type (b).

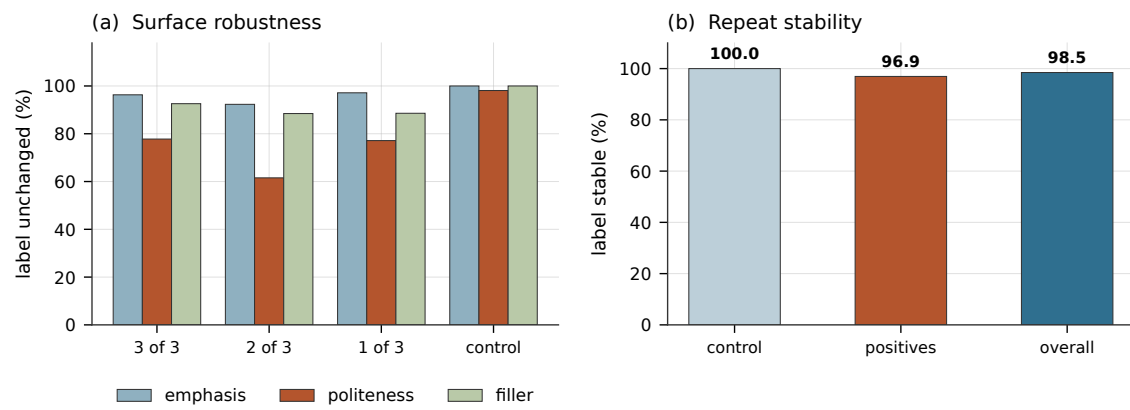


Figure 13. Surface robustness (a) and repeat stability (b), Jev, Codebook A, Condition A.